

PROJECT

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DEPLOY AN APP WITH DOCKER

COMPUTE SERVICE

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Objective

This project aims to containerize a web application using Docker by creating and running a custom image.

It then demonstrates how to deploy the containerized app to AWS Elastic Beanstalk using an IAM user.

Services

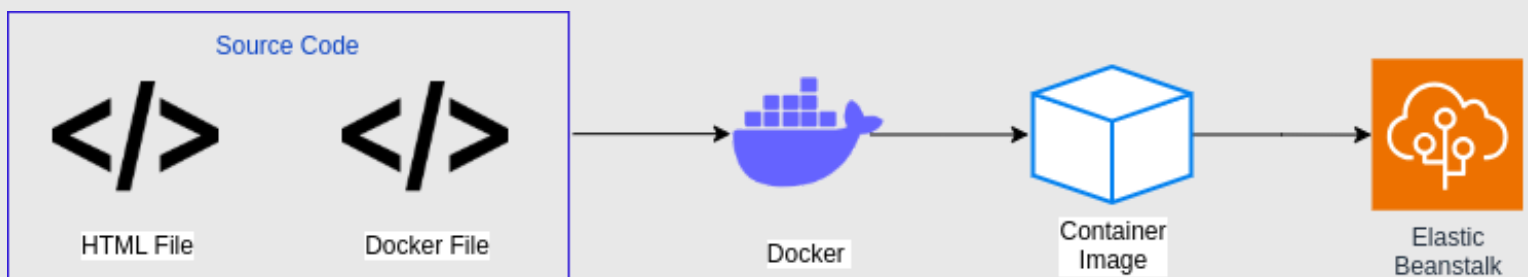
Index.html - The main frontend file that displays the web content to users.

Dockerfile - Defines the environment and steps to build a Docker image for the app.

Docker Container Image - Packages the app with all dependencies for consistent deployment.

Elastic Beanstalk - Automatically deploys and manages the containerized app on AWS.

Architecture Diagram



The index.html file is containerized using a Dockerfile, which builds a Docker image of the application. This image is then uploaded and deployed to AWS using Elastic Beanstalk, which provisions the infrastructure and runs the container for scalable, managed hosting.

Steps to Implement

1. Install Docker on your system:

Download and install Docker from the official Docker website. Verify the installation by running `docker --version` in your terminal.

2. Run a pre-built container image:

Use a command like `docker run hello-world` to test Docker functionality by pulling and running a pre-built image from Docker Hub.

3. Build your custom Docker image:

Create an `index.html` file and a `Dockerfile`. Use the command `docker build -t my-app .` to build a container image with your app.

4. Run a container using your image:

Execute `docker run -p 8080:80 my-app` to start a container and serve your app locally via port 8080.

5. Log in to AWS with your IAM user:

Configure the AWS CLI using `aws configure` with your IAM credentials to enable deployment access from your local environment.

6. Deploy the Docker image to Elastic Beanstalk:

Initialize an Elastic Beanstalk app using `eb init`, create an environment with `eb create`, and deploy your app using `eb deploy`.

Story Time

Docker (The Magic Box Maker):

Docker is your enchanted toolbox that packs your entire app code, ingredients, and spells into a neat little magic box (container) that works the same in every kingdom (system).

index.html (The Front Scroll):

This scroll holds the face of your app the words, buttons, and designs that users see when they open the gate to your magic world.

Dockerfile (The Spellbook):

This is your recipe scroll, where you write out the steps for the box to be built. It tells Docker how to summon the environment, copy the scrolls, and prepare the magic container.

Docker Container Image (The Sealed Potion):

Once brewed, your spell gets bottled into a self-contained potion (image) that can be uncorked and used in any realm whether your own or the cloud kingdom.

AWS IAM (The Royal Key Master):

IAM is your royal key master granting the chosen ones access to portals like Beanstalk while keeping the unworthy out.

Elastic Beanstalk (The Cloud Gatekeeper):

Elastic Beanstalk is the wise old wizard who takes your magic potion and sets it running in the clouds. It grows or shrinks the app's strength depending on how many villagers (users) are knocking at the gates.

BY
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THANK YOU!

